

# Exploring the challenges of Dinoflagellate transcriptome assembly with long-read data

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## Background

Due to climate change, the number of Harmful Algal Blooms (HABs) is increasing. HABs correspond to the rapid and massive proliferation of toxic and/or harmful microalgae. Among them, dinoflagellates are flagellated eukaryotic microorganisms capable of producing toxins. They possess a unique genomic structure, characterized by large DNA content and the absence of histones. Studies have shown that the toxic activity of dinoflagellates is linked to specific transcript expression, making transcriptome analysis particularly relevant.

The aim of this study is to generate a de novo reference transcriptome for three dinoflagellate species using long-read and short-read sequencing data, followed by functional annotation.

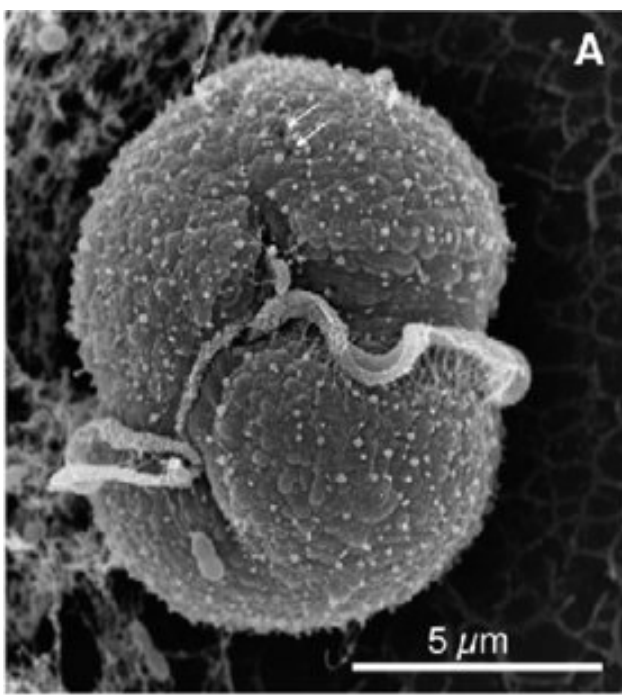
This work is divided into three main steps:

- 1 Identify the most effective cleaning and assembly tools for dinoflagellate transcriptomes using *Prymnesium parvum* as reference.
- 2 Apply these tools to the three dinoflagellate species
- 3 Perform the functional annotation of the assembled transcriptomes.

## Harmful algal bloom (HABs)



*Dinophysis acuminata*



*Karlodinium veneficum*

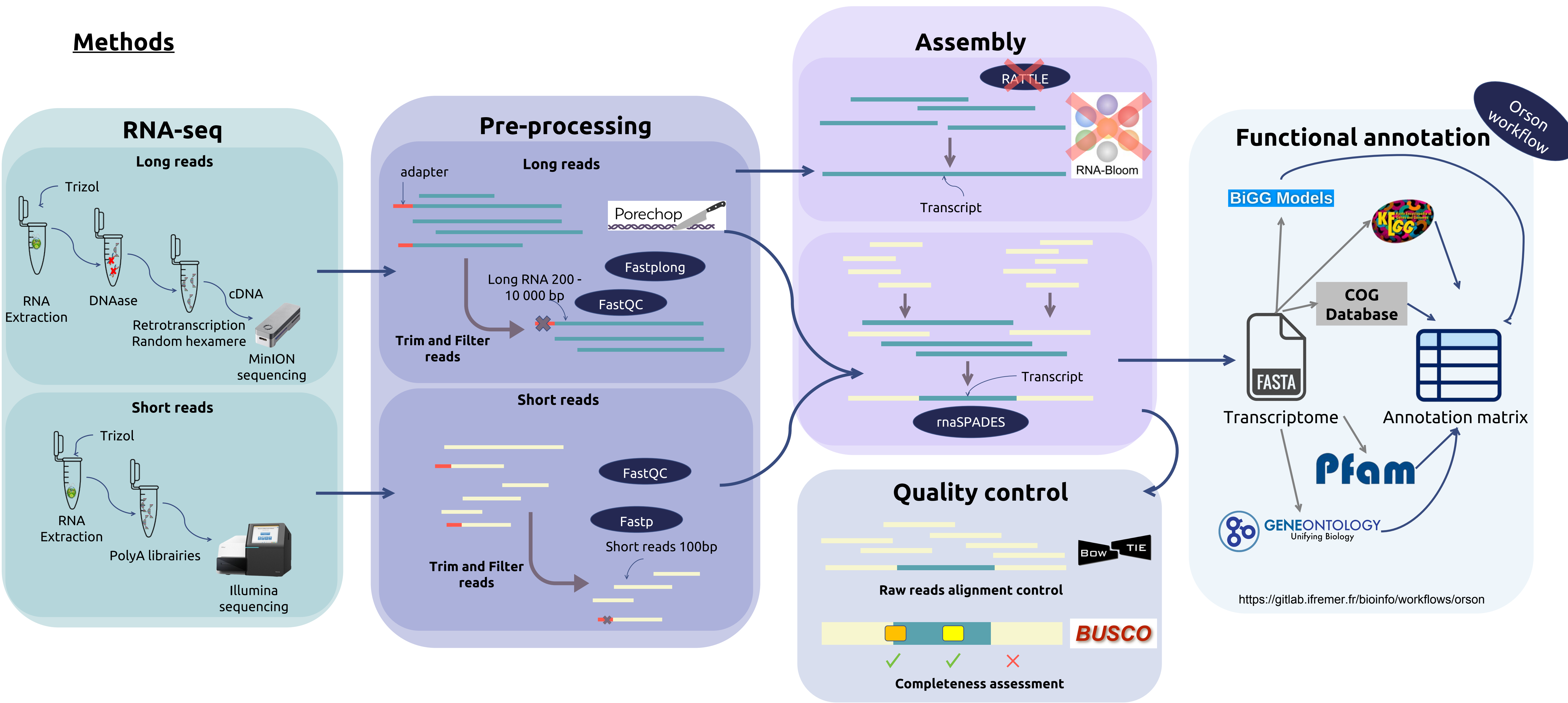


*Lingulodinium polyedrum*



*Prymnesium parvum*  
(Model species with reference genome)

## Methods



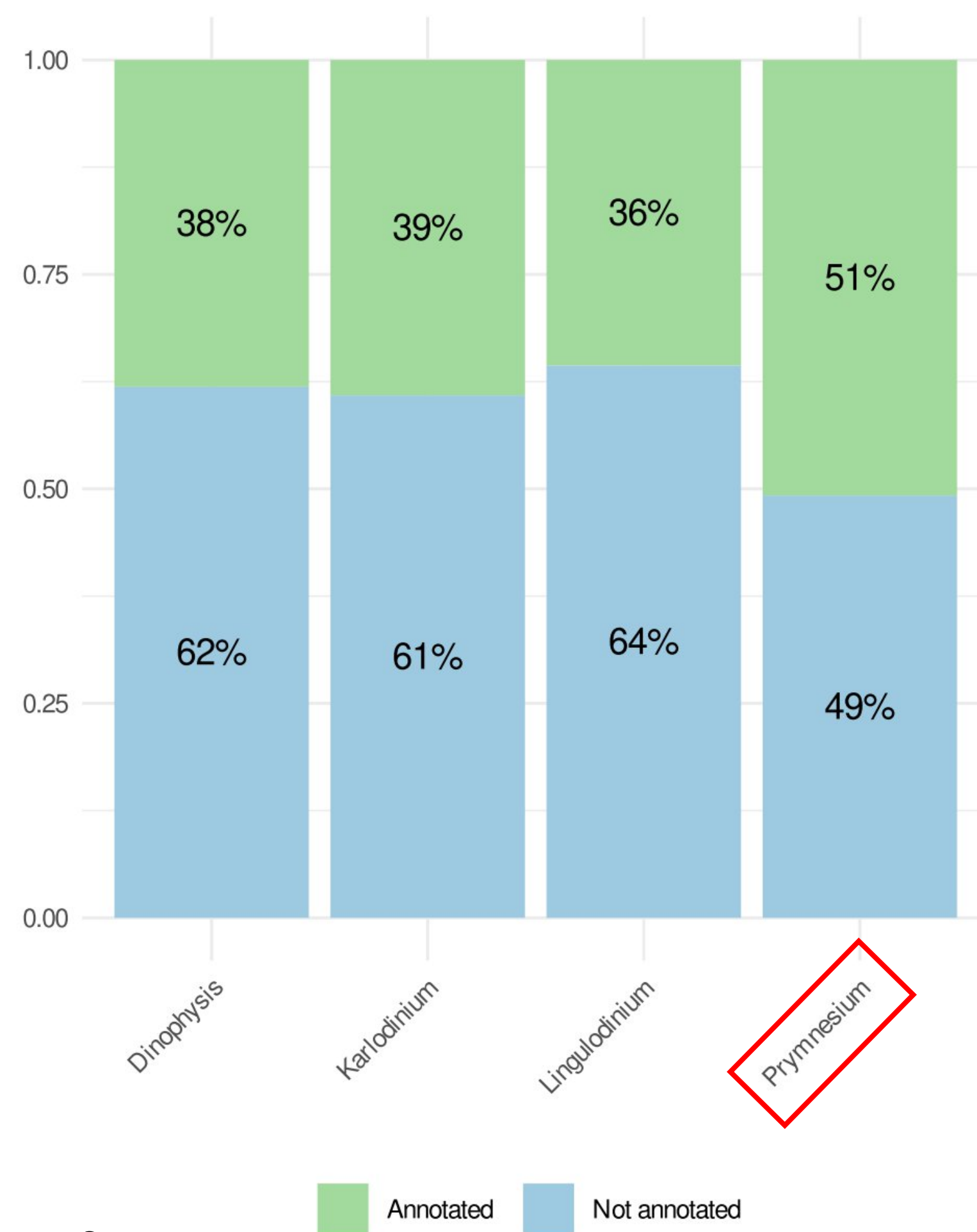
## Results

|                       | Haptophyte<br><i>Prymnesium parvum</i> |                                      | <i>Dinophysis acuminata</i>               |                                      | <i>Karlodinium veneficum</i>              |                                                     | <i>Lingulodinium polyedrum</i> |                                     |
|-----------------------|----------------------------------------|--------------------------------------|-------------------------------------------|--------------------------------------|-------------------------------------------|-----------------------------------------------------|--------------------------------|-------------------------------------|
|                       | <i>RCC3707 strain (Our assembly)</i>   | 121B strain (Wisecaver et al., 2023) | <i>IFR-DAU-01EY strain (Our assembly)</i> | DAEP01 strain (Keeling et al., 2014) | <i>IFR-CC-20-44 strain (Our assembly)</i> | <i>K.micrum CCMP2283 strain (Meng et al., 2018)</i> | <i>IFR (Our assembly)</i>      | CCMP1738 strain (Meng et al., 2018) |
| Assembly type         | Transcriptome                          | Genome                               | Transcriptome                             | Transcriptome                        | Transcriptome                             | Transcriptome                                       | Transcriptome                  | Transcriptome                       |
| Number of transcripts | 84 071                                 | 21 306                               | 112 582                                   | 128 844                              | 136 385                                   | 141 701                                             | 122 606                        | 143 704                             |
| N50                   | 1 986                                  | 6 249                                | 835                                       | 829                                  | 1622                                      | 1 519                                               | 1757                           | 1 386                               |
| BUSCO score           | 90.6 %                                 | 77.30%                               | 45.5%                                     | 41.58%                               | 74.1%                                     | 82.51%                                              | 74.9%                          | 81,85%                              |
| Bowtie2               | 98.58 %                                | -                                    | 85.51%                                    | 47.2%                                | 96.51%                                    | 91.6 %                                              | 97.88%                         | 92,6%                               |

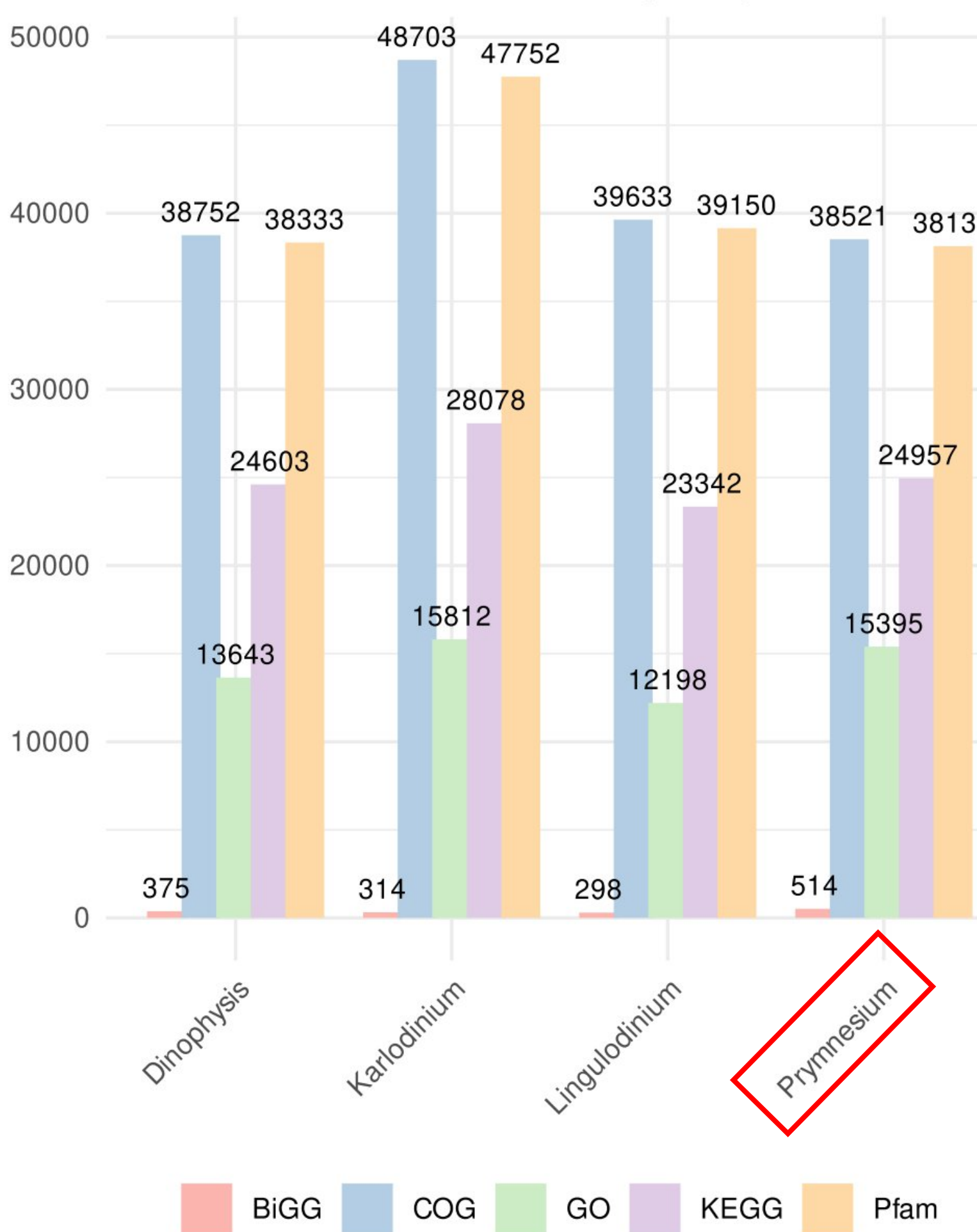
### Comparison with literature:

- All three dinoflagellate species showed slightly higher **alignment score**.
- Lower **BUSCO scores** were observed for *K. veneficum* and *L. polyedrum*, suggesting reduced transcriptome completeness for these species.
- For *P. parvum* the total number of assembled transcripts **greatly exceeded** the expected gene count.

Proportion of annotated transcripts for each species



Number of annotated transcripts by database



### Functional Annotation:

- **Low annotation rates (<40%)** were observed across all three dinoflagellate species.
- *P. parvum* showed a **slightly higher annotation rate**, compared to the other species.

## Conclusion and perspectives

- We successfully generated **transcriptomes** for three dinoflagellate species, despite the known complexity of these non-model organisms.
- In *P. parvum*, the excessive **number of transcripts** compared to its genome suggests possible redundancy or fragmentation introduced by current assembly tools.
  - Direct RNA sequencing may help limit fragmentation and redundancy in future transcriptomic studies
- **BUSCO** and most **functional annotation databases** appear partially suited to these organisms, explaining the low completeness and annotation rates
  - The development of **reference genomes** through initiatives like the **Atlasea Project** will provide resources to enhance annotation accuracy